

# The Modelled Spread

A System of Equations for the COMEX–London Gold Basis

Methods note, GOLD project

16 September 2026

The spread between the COMEX gold futures price and the London benchmark is usually reported as a single annualised number. That number mixes three economically distinct objects: a premium for metal being in New York rather than London, a premium for the risk that a tariff lands before delivery, and the cost of carrying metal through time. This note writes the system down so the three can be held apart, states which combinations the data identify, and records what the estimated versions of them look like.

Equation numbers match Section 9 of `SPREAD_EXPLAINED.html` and Section 4 of `spread_model_note.pdf`. All rates are annualised decimals,  $\tau$  is in calendar days to first notice, and prices are US dollars per troy ounce.

## Notation

| Symbol               | Meaning                                                 | Status                   |
|----------------------|---------------------------------------------------------|--------------------------|
| $S_t$                | London price, unallocated, 15:00 London                 | observed                 |
| $F_t(\tau)$          | COMEX price for delivery in $\tau$ days                 | observed, 13:30 New York |
| $V_t(\tau)$          | value today of metal deliverable in New York at $\tau$  | latent                   |
| $p_t$                | location premium, fraction of spot (place, form, trust) | latent; estimated        |
| $r_t$                | dollar interest rate (SOFR; fed funds before Apr 2018)  | observed                 |
| $s^N$                | New York storage cost, per year                         | assumed; see note below  |
| $\ell_t^N, \ell_t^L$ | lease rate on metal in New York, in London              | latent                   |
| $h_t, \theta_t$      | hazard rate of a tariff; tariff rate if it lands        | latent                   |
| $\Lambda_t(\tau)$    | probability a delivery at $\tau$ is tariffed            | latent                   |
| $\kappa_t$           | all-in cost of moving an ounce between locations        | latent; estimated        |
| $\varepsilon_t$      | gold's return from 15:00 London to the settlement       | latent; measurable       |
| $Q_t$                | metal shipped, tonnes a month                           | observed (Swiss customs) |
| $\bar{Q}$            | refining and secure-freight capacity, tonnes a month    | latent                   |

## The system

**Pricing through time, within New York.**

$$F_t(\tau) = V_t(\tau) \cdot \exp\left[\left(r_t + s^N - \ell_t^N\right) \frac{\tau}{365}\right] \quad (1)$$

**Pricing across locations, and the policy risk inside it.**

$$V_t(\tau) = S_t [1 + p_t + \Lambda_t(\tau) \theta_t] \quad (2)$$

$$\Lambda_t(\tau) = 1 - \exp\left(-h_t \frac{\tau}{365}\right) \approx h_t \frac{\tau}{365} - \frac{1}{2} \left(h_t \frac{\tau}{365}\right)^2 \quad (3)$$

**What arbitrage enforces: a band, not a point.**

$$-\kappa_t^{N \rightarrow L} \leq p_t S_t \leq \kappa_t^{L \rightarrow N} \quad \text{unless capacity binds} \quad (4)$$

$$\kappa_t^{L \rightarrow N} = \underbrace{c_{\text{freight}} + c_{\text{ins}} + c_{\text{recast}}}_{\text{physical, per ounce}} + \underbrace{(r_t + \ell_t^L) \frac{d_{\text{transit}}}{365}}_{\text{financing in transit}} S_t \quad (5)$$

**Measurement: the two legs are not observed at the same instant.**

$$\hat{F}_t(\tau) = F_t(\tau) \cdot e^{\varepsilon_t}, \quad \mathbb{E}[\varepsilon_t] = \mu, \quad \text{Var}[\varepsilon_t] = \sigma^2 \quad (6)$$

The London benchmark is struck at 15:00 London and the COMEX settlement about three and a half hours later, so  $\varepsilon_t$  is gold's return over that window. Every contract settles at the same instant, so  $\varepsilon_t$  is *common* to all of them on date  $t$ : it shifts the whole curve without tilting it.

**Flow, with a capacity ceiling.**

$$Q_t = \min\{\bar{Q}, \beta \max(0, p_t S_t - \kappa_t)\} + u_t \quad (7)$$

**What is estimated**

Taking logs of (1) and (2), substituting (3) and (6), and collecting powers of  $\tau$  gives a curve that is quadratic in the horizon. This is the regression run each day across that date's contracts, weighted by the square root of open interest:

$$\ln \hat{F}_t(\tau_i) = \alpha_t + \beta_t \tau_i + \gamma_t \tau_i^2 + \eta_{it} \quad (8)$$

The three fitted coefficients map onto the objects above:

$$\hat{p}_t \equiv \alpha_t - \ln S_t = p_t + \varepsilon_t \quad (9)$$

$$\hat{\omega}_t \equiv 365\beta_t - r_t = s^N - \ell_t^N + h_t \theta_t \quad (10)$$

$$\gamma_t = -\frac{h_t^2 \theta_t}{2 \cdot 365^2} \leq 0 \quad (11)$$

**The modelled spread**

$$b_t(\tau) \equiv F_t(\tau) - S_t = S_t \left[ \underbrace{p_t}_{\text{place, form, trust}} + \underbrace{\Lambda_t(\tau) \theta_t}_{\text{policy risk}} + \underbrace{(r_t + s^N - \ell_t^N) \frac{\tau}{365}}_{\text{carry}} \right] \quad (12)$$

$$\text{disloc}_t(T) \approx p_t \frac{365}{T} + h_t \theta_t + (s^N - s) - \ell_t^N \quad (13)$$

Equation (12) is the spread itself. Equation (13) is what the project's headline number does to it: reading the fitted curve at a fixed horizon  $T$  and subtracting an assumed carry rate  $s$  divides the location premium by the horizon, leaves the policy and lease terms alone, and adds whatever error sits in the storage assumption. The multiplier  $365/T$  is 12.2 at thirty days, 4.06 at ninety and 2.03 at one hundred and eighty, so the horizon silently sets how much of the reported number is location and how much is carry. That is the case for reporting  $\hat{p}_t$  and  $\hat{\omega}_t$  separately rather than any single dislocation.

## What the system implies, and what the data say

Estimated over 2,857 daily fits, 2015–2026 (`spread_model_checks.py`):

- **The timing error lands in the level, not the slope.** Because  $\varepsilon_t$  is common across contracts it enters (9) and not (10). Regressing  $\hat{p}_t$  on the day's London-to-London return gives +0.193 (s.e. 0.015), against  $3.5/24 = 0.146$  predicted by a random walk over the window; the same regression for  $\hat{\omega}_t$  gives  $-0.012$  (s.e. 0.011), indistinguishable from zero. Any result resting on the slope needs no timing correction.
- **The band in (4) is where the transfer cost is.** In the calm years 2015–19 the fitted level averages  $-\$0.55$  an ounce, so arbitrage does hold  $p_t S_t$  near zero. Trade estimates of the all-in cost of moving London metal into COMEX form are roughly  $\$1.5$ – $\$2.5$  an ounce, and the flow threshold estimated from (7) is  $\$3.77$  with a 90% interval of  $\$0.83$ – $\$10.25$ . Three independent readings of  $\kappa_t$ , mutually consistent.
- **(10) does not identify the lease rate on its own.** The calm-period wedge is  $\hat{\omega} = +0.58$  points a year. With  $\ell^N \approx 0$  in calm years that implies storage of about 0.58%, not the 0.25% the code assumes; if storage really is 0.25%, the implied lease rate is  $-0.33\%$ , which nobody quotes. One of the two is mis-set, and because  $s$  enters (13) one-for-one, so is every reported dislocation level. Levels should be read in differences until this is pinned down.
- **Curvature is the lever on policy risk.**  $\gamma_t$  is negative on average, with mean over standard error of  $-10.7$  in calm years and  $-22.0$  in the tariff window — a roughly sevenfold deepening. Consistent with the hazard term in (3), but present in calm years too, so the dated resolutions (the April 2025 exemption, the July 2025 customs ruling) are still needed to attribute it: at resolution  $h_t \rightarrow 0$ , so the slope and curvature should step while the level does not.
- **The flow equation is not a response function.** Shipments respond to the premium and also close it, so estimating (7) on monthly data recovers an equilibrium relation rather than the supply of arbitrage. The coefficient is descriptive; identification needs the timing within the month, or an event study around the dated announcements.

---

Sources: `claude/spread-explainer/spread_model_checks.py` and `worked_examples.py`, both reading `data/processed/`; the kink threshold is from `claude/mechanism-figures/estimate_kink.py`. Where this note disagrees with earlier project text, it says so.